

Entanglement Detection and Quantification: PT-Moments and Geometric Measure

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Paper Information

Paper 1

Optimal Entanglement Certification from Moments of the Partial Transpose

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Paper 2

Quantifying Entanglement from the Geometric Perspective

Authors: Lisa T. Weinbrenner and Otfried Gühne

Journal: EPL **151**, 68001 (2025)

Outline

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- Partial transpose (PPT)
- Negativity
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- Multipartite entanglement

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Review

Entanglement Detection & Quantification

[Review] Separability & Entanglement

A bipartite state $|\psi\rangle$ is **separable** if it can be written as:

$$|\psi\rangle = \sum_i \lambda_i |\psi_A\rangle |\psi_B\rangle.$$

For density matrix, a bipartite state ρ is **separable** if it can be written as:

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \quad \sum_i p_i = 1, \quad p_i > 0$$

Otherwise, the state is **entangled**.

Approaches to detect entanglement

- **PNCP maps** (e.g., partial transpose): produce negative eigenvalues for entangled states
- **Entanglement witnesses**: observables with non-negative expectation on all separable states

[Review] Partial Transpose & PPT Criterion

For a bipartite state $\rho_{AB} = \sum_{ijkl} \alpha_{ijkl} |i_A\rangle\langle j_A| \otimes |k_B\rangle\langle l_B|$, the **partial transpose** on B is:

$$\rho_{AB}^{T_B} = \sum_{ijkl} \alpha_{ijkl} |i_A\rangle\langle j_A| \otimes |l_B\rangle\langle k_B|$$

which does not change eigen-spectrum for separable state.

PPT Criterion (Peres–Horodecki)

- **Separable** $\Rightarrow \rho^{T_B} \geq 0$ (positive partial transpose)
- **PPT is necessary and sufficient** for 2×2 and 2×3 systems
- In higher dimensions, PPT entangled (bound entangled) states exist

Separable vs. Entangled States

Separable States

For separable states, the partial transpose preserves positivity:

$$\begin{aligned}\rho &= \sum_i p_i \rho_i^A \otimes \rho_i^B \\ &= \sum_i p_i \left(\sum_a \lambda_i^a |\lambda_i^a\rangle \langle \lambda_i^a| \right) \otimes \left(\sum_b \mu_i^b |\mu_i^b\rangle \langle \mu_i^b| \right) \\ &= \sum_{i,a,b} p_i \lambda_i^a \mu_i^b |\lambda_i^a, \mu_i^b\rangle \langle \lambda_i^a, \mu_i^b|\end{aligned}$$

Since all coefficients $p_i \lambda_i^a \mu_i^b \geq 0$ and sum to unity, this represents a valid eigendecomposition. Under positive and trace-preserving operations, the eigenspectrum remains positive.

Separable vs. Entangled States

Entangled States

For entangled states, quasi-probability decomposition becomes necessary:

$$\rho = \sum_i q_i |a_i, b_i\rangle \langle a_i, b_i|$$

where $\sum_i q_i = 1$ but some $q_i < 0$.

The states $\{|a_i, b_i\rangle\}$ form an overcomplete (non-orthonormal) basis. Under PNCP operations, while the quasi-probability coefficients remain unchanged, the transformation of the overcomplete basis induces negative eigenvalues.

[Review] Negativity & Logarithmic Negativity

The **negativity** quantifies the violation of PPT:

$$\mathcal{N}(\rho) = \frac{\|\rho^{T_B}\| - 1}{2} = \sum_{\lambda_i < 0} |\lambda_i|$$

where $\|A\| = \text{Tr}\sqrt{A^\dagger A}$ is the trace norm.

The **logarithmic negativity** is defined as:

$$E_N(\rho_{AB}) = \ln \|\rho_{AB}^{T_B}\|$$

Properties:

- Separable states: $\|\rho^{T_B}\| = 1 \Rightarrow E_N = 0$
- Entangled states (detected by PPT): $\|\rho^{T_B}\| > 1 \Rightarrow E_N > 0$
- For the Bell state: eigenvalues of ρ^{T_B} are $\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$, giving $E_N = \ln 2$

[Review] Entanglement Witnesses

An observable W is an **entanglement witness** if:

- $\text{Tr}(W\rho) \geq 0$ for all separable states ρ
- $\text{Tr}(W\rho) < 0$ for some entangled state ρ

Decomposable Witnesses

A witness W is **decomposable** w.r.t. bipartition $M|\bar{M}$ if:

$$W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0$$

A witness is **fully decomposable** if this holds for **all** bipartitions M .

Key theorem: Fully decomposable witnesses detect **exactly** the states that are **not PPT mixtures**:

$$\rho \notin \mathcal{S}_{\text{PPT-mix}} \implies \exists W \in \mathcal{W}_{\text{fd}} : \text{Tr}(W\rho) < 0$$

[Review] Multipartite Entanglement

A state is **biseparable** if it can be written as:

$$\rho^{\text{bs}} = \sum_i p_i \rho_{M_i|\bar{M}_i}^{\text{sep}}, \quad \sum_i p_i = 1$$

A state has **genuine multipartite entanglement (GME)** iff it is NOT biseparable.

GME Detection via SDP

$$\begin{aligned} & \min_{W, P_M, Q_M} \quad \text{Tr}(W\rho) \\ & \text{subject to} \quad \text{Tr}(W) = 1 \\ & \quad \quad \quad W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0 \quad \forall M \end{aligned}$$

If $\min \text{Tr}(W\rho) < 0$, the state is **GME**.

Limitation: All these methods generally require **full density matrix** ρ — expensive quantum state tomography, and even numerical search for entanglement witness W .

[Review] Summary

Method	Detects	Requires	Limitation
PPT	Entanglement (NPT states)	ρ^{T_B} spectrum	Misses PPT entangled states
Negativity	+ Quantifies	ρ^{T_B} eigenvalues	Not a full measure
Witnesses	Tailored detection	Observable $\mathcal{W}$	Must construct $\mathcal{W}$ per state
SDP (GME)	Genuine multipartite	Full ρ	Exponential scaling

Question: Can we certify entanglement **without full state tomography**?

⇒ **Paper 1:** Use only the *moments* of ρ^{T_A} — accessible via randomized measurements.

Optimal Entanglement Certification from Moments of the Partial Transpose

Yu, Imai, and Gühne (PRL 2021)

Motivation: Why PT-Moments?

Challenge: Full state tomography scales exponentially. Can we detect entanglement from **partial** information?

PT-Moments

The n -th moment of the partially transposed state is:

$$p_n = \text{Tr} \left[(\rho_{AB}^{T_A})^n \right]$$

Key advantage: PT-moments can be efficiently measured via **randomized measurements** (classical shadows, U -statistics) without reconstructing ρ .

PT-Moments: Definition and Properties

Given a bipartite state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = d$, the **PT-moments** are:

$$p_n = \text{Tr}[(\rho_{AB}^{T_A})^n], \quad n = 0, 1, 2, \dots$$

Basic properties

- $p_0 = d$ (dimension)
- $p_1 = \text{Tr}[\rho_{AB}^{T_A}] = \text{Tr}[\rho_{AB}] = 1$
- $p_2 = \text{Tr}[(\rho_{AB}^{T_A})^2] = \text{Tr}[\rho_{AB}^2]$ (purity), so $\frac{1}{d} \leq p_2 \leq 1$
- For **separable** states: $\rho_{AB}^{T_A} \geq 0$, so $p_n = \text{Tr}[(\rho_{AB}^{T_A})^n] \geq 0$ for all n , and the spectrum $(x_1, \dots, x_d)$ satisfies $x_i \geq 0$

The PT-Moment Problem

Given moments $(p_1, p_2, \dots, p_n)$, is there a **separable state** ρ_{AB} with these PT-moments?

If not $\Rightarrow$ the state **must be entangled**.

PT-Moments: Eigen-spectrum

If we are aware of all the PT-moments $p = (p_1, p_2, \dots, p_d)$ where $d = d_A d_B$, all the eigenvalues of ρ_{AB}^T can be directly calculated. It is just solving equation, employing Newton's identities.

$$p_1 = \text{Tr} \rho_{AB}^T = \sum_i \lambda_i$$

$$p_2 = \text{Tr} \rho_{AB}^{T^2} = \sum_i \lambda_i^2$$

$$p_3 = \text{Tr} \rho_{AB}^{T^3} = \sum_i \lambda_i^3$$

$$\vdots \quad \quad \quad \vdots$$

PT-Moments: Eigen-spectrum

The eigenvalues $\{\lambda_i\}$ are roots of

$$P(x) = \prod_{i=1}^d (x - \lambda_i) = x^d - e_1 x^{d-1} + e_2 x^{d-2} - \dots + (-1)^d e_d$$

where e_k are the **elementary symmetric polynomials**:

$$e_1 = \sum_i \lambda_i, \quad e_2 = \sum_{i < j} \lambda_i \lambda_j, \quad e_3 = \sum_{i < j < k} \lambda_i \lambda_j \lambda_k, \quad \dots$$

These provide a recursive relation between $s_k = \sum_i \lambda_i^k$ and e_k : $s_1 = e_1$, $s_2 = s_1 e_1 - 2e_2$, $s_3 = s_2 e_1 - s_1 e_2 + 3e_3$, $\dots$, and in general:

$$s_k = \sum_{j=1}^{k-1} (-1)^{j-1} s_{k-j} e_j + (-1)^{k-1} k e_k$$

PT-Moments: Eigen-spectrum

For example of two-qubit system,

$$e_1 = 1$$

$$e_2 = \frac{p_1^2 - p_2}{2} = \frac{1 - p_2}{2}$$

$$e_3 = \frac{p_1^3 - 3p_1p_2 + 2p_3}{6} = \frac{1 - 3p_2 + 2p_3}{6}$$

$$e_4 = \frac{p_1^4 - 6p_1^2p_2 + 3p_2^2 + 8p_1p_3 - 6p_4}{24} = \frac{1 - 6p_2 + 3p_2^2 + 8p_3 - 6p_4}{24}$$

The eigenvalues of $\rho_{AB}^{T_A}$ are then the four roots of

$$x^4 - e_1x^3 + e_2x^2 - e_3x + e_4 = 0.$$

Connection to Classical Moment Problems

Writing the spectrum of $\rho_{AB}^{T_A}$ as $(x_1, x_2, \dots, x_d)$, we have:

$$p_n = \sum_{i=1}^d x_i^n$$

This connects the PT-moment problem to **classical moment problems**:

Hamburger Moment Problem: Given moments $m^{(n)} = (m_0, m_1, \dots, m_n)$, does there exist a measure on $\mathbb{R}$ with these moments?

Stieltjes Moment Problem: Same question, but the measure is supported on $[0, \infty)$.

For **separable** states, $x_i \geq 0 \Rightarrow$ Stieltjes problem.

For **general** (possibly entangled) states, $x_i \in \mathbb{R} \Rightarrow$ Hamburger problem.

The key mathematical tool for solving both: **Hankel matrices**.

Why Hankel Matrices?

Gram matrix construction (Appendix A)

Use the Hilbert-Schmidt inner product $(X, Y) := \text{Tr}(X^\dagger Y)$ in operator space.

Given $\rho \geq 0$ and $X \geq 0$ satisfying $m_k = \text{Tr}(\rho X^k) = \langle \varphi | X^k | \varphi \rangle$, construct the operator sequences:

$$v = \left(\rho^{1/2}, \rho^{1/2} X, \dots, \rho^{1/2} X^{\lfloor n/2 \rfloor} \right), \quad u = \left(\rho^{1/2} X^{1/2}, \rho^{1/2} X^{3/2}, \dots, \rho^{1/2} X^{\lfloor \frac{n-1}{2} \rfloor + 1/2} \right)$$

Their Gram matrices are:

$$(v_i, v_j) = \text{Tr}(X^i \rho X^j) = \text{Tr}(\rho X^{i+j}) = m_{i+j} \quad \Rightarrow \quad [H_k]_{ij} = m_{i+j}$$

$$(u_i, u_j) = \text{Tr}(X^{i+1/2} \rho X^{j+1/2}) = m_{i+j+1} \quad \Rightarrow \quad [B_k]_{ij} = m_{i+j+1}$$

Since Gram matrices are always positive semidefinite, we get:

$$\rho \geq 0 \Rightarrow H_{\lfloor n/2 \rfloor} \geq 0, \quad \rho \geq 0 \text{ and } X \geq 0 \Rightarrow B_{\lfloor (n-1)/2 \rfloor} \geq 0$$

Hankel Matrix Criterion

Define the **Hankel matrices** from a moment sequence $m^{(n)} = (m_0, m_1, \dots, m_n)$:

$$H_k = \begin{pmatrix} m_0 & m_1 & \cdots & m_k \\ m_1 & m_2 & \cdots & m_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ m_k & m_{k+1} & \cdots & m_{2k} \end{pmatrix}, \quad B_k = \begin{pmatrix} m_1 & m_2 & \cdots & m_{k+1} \\ m_2 & m_3 & \cdots & m_{k+2} \\ \vdots & \vdots & \ddots & \vdots \\ m_{k+1} & m_{k+2} & \cdots & m_{2k+1} \end{pmatrix}$$

Lemma (Classical moment theory)

(a) $m^{(n)} \in \mathcal{M}_n$ (Hamburger, measure on $\mathbb{R}$) $\iff H_{\lfloor n/2 \rfloor} \geq 0$ where

$$\mathcal{M}_n = \left\{ m^{(n)} \mid \text{Tr}(\sigma X^k) = m_k, \sigma \geq 0, X^\dagger = X \right\}$$

(b) $m^{(n)} \in \mathcal{M}_n^+$ (Stieltjes, measure on $[0, \infty)$) $\iff H_{\lfloor n/2 \rfloor} \geq 0$ and $B_{\lfloor (n-1)/2 \rfloor} \geq 0$

$$\mathcal{M}_n = \left\{ m^{(n)} \mid \text{Tr}(\sigma X^k) = m_k, \sigma \geq 0, X \geq 0 \right\}$$

Since separable states have $x_i \geq 0$, the Stieltjes condition applies. Violation of either Hankel condition certifies entanglement. Let $\sigma = I, X = \rho_{AB}^T$.

p_n -PPT Criterion

Theorem (Entanglement criteria from Hankel matrices)

Let $p_k = \text{Tr}[(\rho_{AB}^{T_A})^k]$ for $k = 1, 2, \dots, n$. A necessary condition for ρ_{AB} being separable is:

$$H_{\lfloor n/2 \rfloor} \geq 0 \quad \text{and} \quad B_{\lfloor (n-1)/2 \rfloor} \geq 0$$

Lowest order: p_3 -PPT criterion

For $n = 3$, the Hankel condition $H_1 \geq 0$ gives:

$$H_1 = \begin{pmatrix} p_1 & p_2 \\ p_2 & p_3 \end{pmatrix} = \begin{pmatrix} 1 & p_2 \\ p_2 & p_3 \end{pmatrix} \geq 0$$

This requires $\det(H_1) \geq 0$, yielding the p_3 -PPT criterion:

$$\boxed{p_3 \geq p_2^2}$$

If this is violated, the state is **entangled**. Higher orders ($n = 5, 7, \dots$) give strictly stronger criteria.

Optimal Solution to the PT-Moment Problem

The Hankel criteria are **not optimal** — we may assume $\sigma \neq I$.

Exact optimization formulation

For fixed $(p_1, p_2, \dots, p_{n-1})$, find the range of p_n compatible with a separable state:

$$\begin{aligned} \min_{x_i} / \max_{x_i} \quad & \hat{p}_n := \sum_{i=1}^d x_i^n \\ \text{subject to} \quad & \sum_{i=1}^d x_i^k = p_k, \quad k = 1, 2, \dots, n-1 \\ & x_i \geq 0 \quad \text{for } i = 1, 2, \dots, d \end{aligned}$$

The solutions are found **analytically** using Cramer's rule and the Vandermonde determinant. The extremal spectra have a specific structure with at most **three distinct values**.

p_3 -OPPT Criterion (Theorem 3)

Theorem (Optimal p_3 -OPPT)

(a) There exists a d -dimensional **separable (PPT)** state ρ_{AB} satisfying $p_k = \text{Tr}[(\rho_{AB}^{T_A})^k]$ for $k = 1, 2, 3$ if and only if:

$$p_1 = 1, \quad \frac{1}{d} \leq p_2 \leq 1$$

$$\alpha x^3 + (1 - \alpha x)^3 \leq p_3 \leq [1 - (d - 1)y]^3 + (d - 1)y^3$$

where:

$$\alpha = \left\lfloor \frac{1}{p_2} \right\rfloor, \quad x = \frac{\alpha + \sqrt{\alpha[p_2(\alpha + 1) - 1]}}{\alpha(\alpha + 1)}, \quad y = \frac{d - 1 - \sqrt{(d - 1)(p_2 d - 1)}}{d(d - 1)}$$

(b) If the p_k for $k = 1, 2, 3$ are PT-moments from a **quantum state**, they are compatible with a separable state if and only if:

$$\boxed{p_3 \geq \alpha x^3 + (1 - \alpha x)^3}$$

The p_3 -OPPT criterion is **dimension-independent** and strictly stronger than p_3 -PPT.

Comparison: p_3 -PPT vs p_3 -OPPT

Hierarchical structure

The criteria form a hierarchy based on PT-moments:

- p_3 -PPT ($n = 3$, Hankel): $p_3 \geq p_2^2$
- p_3 -OPPT ($n = 3$, optimal): $p_3 \geq \alpha x^3 + (1 - \alpha x)^3$
- p_n -PPT ($n = 3, 5, 7, \dots$): higher-order Hankel conditions
- p_n -OPPT ($n = 3, 5, 7, \dots$): higher-order optimal conditions

Detection Power

Fraction of entangled $D \times D$ states detected (Hilbert–Schmidt distribution)

D	NPT	NPT3	NPT5	ONPT3	ONPT4	NPT5
2	75.68%	25.53%	39.97%	75.68%	64.78%	75.68%
3	99.99%	25.32%	39.46%	91.63%	97.51%	98.97%
4	100%	23.29%	33.69%	98.68%	100%	100%
5	100%	21.80%	34.54%	99.95%	100%	100%
6	100%	20.93%	31.20%	100%	100%	100%

- **NPT**: PPT criterion (full spectrum)
- **NPT n** : p_n -PPT criterion
- **ONPT n** : p_n -OPPT criterion (optimal)

The optimal criterion ONPT3 detects **significantly more** states than NPT3, and approaches full NPT detection power. ONPT4 and NPT5 achieve near-complete detection.

Summary: PT-Moment Certification

1. **Hankel matrix method** (p_n -PPT): relaxation to classical moment problems
 - Lowest order gives $p_3 \geq p_2^2$
 - Higher orders provide strictly stronger criteria
2. **Optimal method** (p_n -OPPT): exact solution of the PT-moment problem
 - Gives necessary and sufficient conditions
 - Detects significantly more entangled states
 - Dimension-independent
3. **Advantages** over conventional methods
 - **No full tomography required** — only PT-moments from randomized measurements
 - **No prior information** needed (unlike witness-based methods)
 - **Efficient measurement scaling** — higher-order moments cost only marginally more

Quantifying Entanglement from the Geometric Perspective

Weinbrenner and Gühne (EPL 2025)

Geometric Measure of Entanglement

The **geometric measure** quantifies entanglement by the distance to the nearest separable (product) state.

Definition: For a pure state $|\psi\rangle$ of N particles, define the **maximal overlap** with product states:

$$\Lambda^2(\psi) = \max_{|\phi\rangle=|a_1\rangle\otimes|a_2\rangle\otimes\cdots\otimes|a_N\rangle} |\langle\phi|\psi\rangle|^2$$

The **geometric measure of entanglement** is $E_G(\psi) = 1 - \Lambda^2(\psi)$.

Schmidt Decomposition and Λ

Bipartite case

Any bipartite state can be written in the **Schmidt decomposition**:

$$|\psi\rangle = \sum_{i=1}^r s_i |a_i\rangle |b_i\rangle$$

where $s_1 \geq s_2 \geq \dots \geq s_r > 0$ are the **Schmidt coefficients**, $\sum_i s_i^2 = 1$, and r is the **Schmidt rank**.

The maximal overlap is simply the **largest Schmidt coefficient** $\Lambda(\psi) = s_1$.

The Schmidt decomposition follows from the SVD of the coefficient matrix τ_{ij} :

$$|\psi\rangle = \sum_{i,j} \tau_{ij} |i\rangle_A |j\rangle_B$$

A state is separable iff the Schmidt rank $r = 1$ (i.e., τ_{ij} has rank one and must be one).

Basic Examples

GHZ state

$$|\text{GHZ}_N\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N})$$

$$\Lambda^2 = \frac{1}{2}, \quad E_G = \frac{1}{2}$$

Nearest product state: $|0\rangle^{\otimes N}$ or $|1\rangle^{\otimes N}$, independent of N .

W state

$$|W_N\rangle = \frac{1}{\sqrt{N}}(|10\cdots 0\rangle + |01\cdots 0\rangle + \cdots + |0\cdots 01\rangle)$$

$$\Lambda^2 = \left(\frac{N-1}{N}\right)^{N-1} \xrightarrow{N \rightarrow \infty} \frac{1}{e}, \quad E_G = 1 - \Lambda^2$$

Computation and the Injective Tensor Norm

Multipartite states

For an N -partite state $|\psi\rangle = \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} |i_1 \dots i_N\rangle$, the maximal overlap is:

$$\Lambda(\psi) = \max_{|a_k\rangle} \left| \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} (a_1)_{i_1} \dots (a_N)_{i_N} \right|$$

Injective tensor norm

This is precisely the **injective tensor norm** of the coefficient tensor T :

$$\|T\|_{\sigma} = \max_{\|a_k\|=1} |T(a_1, a_2, \dots, a_N)| = \Lambda(\psi)$$

The injective tensor norm is also called the **spectral norm** of the tensor.

Remark: Computing $\Lambda(\psi)$ for multipartite states is in general **NP-hard**.

Why Tensor Eigenvalues?

The computational problem

Computing the geometric measure requires solving:

$$\Lambda(\psi) = \max_{\|a_k\|=1} \left| \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} (a_1)_{i_1} \cdots (a_N)_{i_N} \right|$$

This is a **polynomial optimization on the product of unit spheres** — a non-convex problem.

Why Tensor Eigenvalues?

Why eigenvalues?

For **matrices** ($N = 2$), the analogous problem $\max_{\|x\|=1} |x^T A y|$ is solved by the **singular value decomposition**. The critical points satisfy $Ay = \sigma x$, $A^T x = \sigma y$.

For **tensors** ($N \geq 3$), no SVD exists in general. Instead, the critical points of the optimization satisfy a **tensor eigenvalue equation**:

$$T\bar{x}^{N-1} = \lambda x, \quad \|x\| = 1$$

The **largest eigenvalue** gives $\Lambda(\psi)$. Tensor eigenvalue theory thus provides the natural mathematical framework for characterizing the extrema of the geometric measure.

Tensor Eigenvalues: Setup

The coefficient tensor $T_{i_1 i_2 \dots i_N}$ of a quantum state $|\psi\rangle$ defines a multilinear form. The geometric measure connects to the **eigenvalue theory of tensors**.

Tensor notation

For a tensor T of order N with dimensions $d_1 \times d_2 \times \dots \times d_N$, define the contraction:

$$T\bar{x}^{N-1} := \sum_{i_2, \dots, i_N} T_{i_1 i_2 \dots i_N} \bar{x}_{i_2}^{(2)} \dots \bar{x}_{i_N}^{(N)}$$

Eigenvalue equation

A scalar λ and vector x satisfy the **tensor eigenvalue equation** if:

$$T\bar{x}^{N-1} = \lambda x$$

with the normalization $x^T x = 1$ (or $x^\dagger x = 1$ in the complex case).

The **largest eigenvalue** of the tensor equals $\Lambda(\psi)$, the maximal overlap with product states.

Z-Eigenvalues

Definition (for real symmetric tensors)

A scalar $\lambda \in \mathbb{R}$ is a **Z-eigenvalue** of a real symmetric tensor T if there exists $x \in \mathbb{R}^d$ with $x^T x = 1$ such that:

$$T\bar{x}^{N-1} = \lambda x$$

Properties

- Z-eigenvalues always exist for symmetric tensors
- Every Z-eigenvalue is a **critical value** of the polynomial $T(x, x, \dots, x)$ on the unit sphere
- The **largest Z-eigenvalue** equals $\|T\|_\sigma$ (the injective tensor norm)

US-Eigenvalues

Beyond real tensors: unitary eigenvalues

For complex tensors (general quantum states), Z-eigenvalues are insufficient. The appropriate generalization uses **US-eigenvalues** (unitary-similarity eigenvalues).

Definition

A scalar λ is a **US-eigenvalue** of tensor T if there exist unitaries $U_2, \dots, U_N$ and a vector x with $x^\dagger x = 1$ such that:

$$T(I, U_2, \dots, U_N) \bar{x}^{N-1} = \lambda x$$

Key result

For a general N -partite state $|\psi\rangle$ with coefficient tensor T :

$$\Lambda(\psi) = \max\{|\lambda| : \lambda \text{ is a US-eigenvalue of } T\}$$

Asymptotic Behaviour of Generic States

Key result

For Haar-random N -qubit pure states ($N > 11$):

$$\text{Prob}(\Lambda^2(\psi) > 3N^2 \cdot 2^{-N}) < \exp(-N)$$

This means that for generic multi-qubit states, the maximal overlap with product states is **exponentially small**:

$$\Lambda^2(\psi) \lesssim 3N^2 \cdot 2^{-N}$$

Physical interpretation

- The geometric entanglement $E_G = 1 - \Lambda^2 \rightarrow 1$ for almost all states
- Generic states are highly entangled in the geometric sense

Mixed State Extension: Convex Roof

For mixed states ρ , the geometric measure might be extended via the convex roof construction:

$$\Lambda^2(\rho) = \max_{\{p_i, |\psi_i\rangle\}} \sum_i p_i \Lambda^2(|\psi_i\rangle) = \min_{\{p_i, |\psi_i\rangle\}} \sum_i p_i (1 - \Lambda^2(|\psi_i\rangle))$$

where the minimization is over all pure-state decompositions $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$.

The convex roof minimization is generally intractable. Analytical results exist only for special cases (e.g., permutation-symmetric states).

Summary: The Geometric Perspective

Concepts

- Injective tensor norm: $= \Lambda(\psi)$, maximal product state overlap
- Z-eigenvalues: Extrema on real unit sphere $\rightarrow \Lambda$ for real states
- US-eigenvalues: Complex generalization $\rightarrow \Lambda$ for arbitrary states

Open directions

- Extending the geometric measure to multipartite mixed states efficiently
- Connecting tensor rank to operational entanglement properties
- Better algorithms for computing the injective tensor norm

Thank You!

Questions & Discussion